

Intrinsic Surface Properties from Surface Triangulation

Xin CHEN and Francis SCHMITT

Ecole Nationale Supérieure des Télécommunications
46 Rue Barrault 75013 PARIS - FRANCE

1 Introduction

Intrinsic surface properties are those properties which are not affected by the choice of the coordinate system, the position of the viewer relative to the surface, and the particular parameterization of the surface. In [2], Besl and Jain have argued the importance of the *surface curvatures* as such intrinsic properties for describing the surface. But such intrinsic properties may be useful only when they can be stably computed. Most of the techniques proposed so far for computing surface curvatures can only be applied to range data represented in image form (see [5] and references therein). But in practice, it is not always possible to represent the sampled data under this form, as in the case of closed surfaces. So other representations must be used.

Surface triangulation refers to a computational structure imposed on the set of 3D points sampled from a surface to make explicit the proximity relationships between these points [1]. Such structure has been used to solve many problems [1]. One question concerning such structure is what properties of the underlying surface can be computed from it. It is obvious that some geometric properties, such as area, volume, axes of inertia, surface normals at the vertices, can be easily estimated [1]. But it is less clear how to compute some other intrinsic surface properties. In [8], a method for computing the minimal (geodesic) distance on a triangulated surface has been proposed. Lin and Perry [6] have discussed the use of surface triangulation to compute the Gaussian curvature and the genus of surface. In this paper, we propose a scheme for computing the principal curvatures at the vertices of a triangulated surface.

2 Principal Curvatures from Surface Triangulation

The basic recipe of the computation is based on the Meusnier and the Euler theorem. We will firstly describe how to use them to compute the principal curvatures. Then the concrete application of the idea to surface triangulation is presented. Throughout we take $\|\bullet\|$ as the L^2 norm, and $\langle \bullet, \bullet \rangle$ and $\wedge$ the inner and cross product, respectively.

2.1 Computing Principal Curvatures by Meusnier and Euler Theorem

Let $\mathbf{N}$ be the unit normal to a surface S at point P . Given a unit vector $\mathbf{T}$ in the *tangent plane* to S at P , we can pass through P a curve $C \subset S$ which has $\mathbf{T}$ as its tangent vector at P . Now let κ be the curvature of C at P , and $\cos\theta = \langle \mathbf{n}, \mathbf{N} \rangle$, where $\mathbf{n}$ is the normal vector to C at P (see Fig. 1). The number

$$\kappa_{\mathbf{T}} = \kappa \cos\theta \quad (1)$$

is called the *normal curvature* of C at P . Note that the sign of the normal curvature of C changes with the choice of the orientation of surface normal $\mathbf{N}$. The Meusnier theorem states that all curves lying on S and having the same tangent vector $\mathbf{T}$ at P have at

this point the same normal curvature [4]. Among all these curves, a particular one is the *normal section* of S at P along $\mathbf{T}$, which is obtained by intersecting S with a plane containing $\mathbf{T}$ and $\mathbf{N}$ (see Fig. 1). For this curve, its normal $\mathbf{n}$ is aligned with $\mathbf{N}$ but with the same or an opposite orientation. Thus from equation (1), its curvature satisfies the expression $\kappa = |\kappa_{\mathbf{T}}|$.

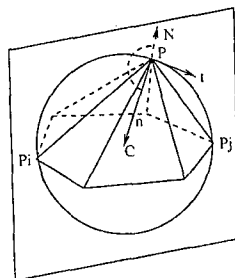
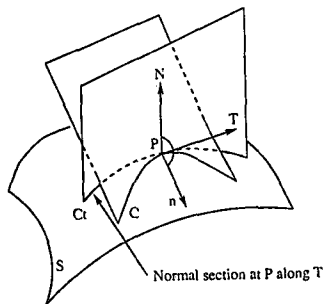


Fig. 1. Local surface geometry around point P .

Fig. 2. Choice of vertex triples.

If we let the unit vector $\mathbf{T}$ rotate around $\mathbf{N}$, we can define an infinite number of normal sections, each of which is associated with a normal curvature $\kappa_{\mathbf{T}}$. Among them, there are two sections, which occur in orthogonal directions, whose normal curvature attains maximum and minimum, respectively [4]. These two normal curvatures are the *principal curvatures* κ_1 and κ_2 , their associated directions are the *principal directions* $\mathbf{T}_1$ and $\mathbf{T}_2$. The Euler theorem gives the relation between the normal curvature $\kappa_{\mathbf{T}}$ of an arbitrary normal section $\mathbf{T}$ and κ_1, κ_2 as follows [4]:

$$\kappa_{\mathbf{T}} = \kappa_1 \cos^2 \Phi + \kappa_2 \sin^2 \Phi, \quad (2)$$

where Φ is the angle between $\mathbf{T}$ and $\mathbf{T}_1$. Let

$$\xi = \frac{\cos \Phi}{(|\kappa_{\mathbf{T}}|)^{1/2}}, \quad \eta = \frac{\sin \Phi}{(|\kappa_{\mathbf{T}}|)^{1/2}}. \quad (3)$$

Then relation (2) becomes

$$\kappa_1 \xi^2 + \kappa_2 \eta^2 = \pm 1, \quad (4)$$

where the sign of the right hand depends on the choice of orientation of the normal $\mathbf{N}$ at P . Equation (4) defines what is known as the *Dupin indicatrix* of the surface S at P . We see that the Dupin indicatrix is a conic defined by κ_1 and κ_2 in the tangent plane to S at P . If P is an elliptic point, the Dupin indicatrix is an ellipse (κ_1 and κ_2 have the same sign). If P is a hyperbolic point, κ_1 and κ_2 have opposite signs, thus the Dupin indicatrix is made up of two hyperbola. If axes other than those in the directions of principal curvatures are used, the Dupin indicatrix would take the following general form:

$$A\xi^2 + 2B\xi\eta + C\eta^2 = \pm 1, \quad (5)$$

Given these two theorems, a possible scheme to calculate the principal curvatures is as follows: Let n plane curves (not necessarily normal sections) passing through the point P be given. For each of them, we can compute its curvature and tangent vector (and thus its normal vector) at P . From the n computed tangent vectors, the surface normal at P

can be determined by applying a vector product to two of these tangent vectors. Using equation (1), the normal curvatures along n tangent directions are computed. Having chosen two orthogonal axes on the tangent plane to S at P , we use equation (3) to compute a pair of coordinates (ξ, η) for each direction (note that this time Φ is the angle between the tangent vector and one of the chosen axes), thus we obtain an equation (5). With n ($n \geq 3$) such equations, the three unknowns A , B , and C can be solved. Finally, the principal curvatures κ_1 and κ_2 are

$$\kappa_1 = \frac{1}{2}[A + C + \sqrt{(A - C)^2 + 4B^2}], \quad \kappa_2 = \frac{1}{2}[A + C - \sqrt{(A - C)^2 + 4B^2}]. \quad (6)$$

The principal directions are determined by performing a rotation of the two orthogonal axes, in the tangent plane, by an angle ψ where $\tan 2\psi = 2B/(A - C)$.

2.2 Principal Curvatures from Surface Triangulation

Suppose that a surface triangulation has been obtained. This triangulation connects each vertex P to a set N_P of vertices, which are the surface neighbors of P in different directions. It is to be noted that such a neighborhood relationship is not only a topological one, but also a geometric one. Our goal is to calculate the principal curvatures of the underlying surface at P using this neighborhood relationship.

We see from the above section that if a set of surface curves passing through P can be defined, the calculation of the principal curvatures and directions will be accomplished by simply invoking the Meusnier and the Euler theorem. So the problem is reduced to define such a set of surface curves. As the neighborhood of P defined by the triangulation reflects the local surface geometry around P , it is natural to define the surface curves from the vertices in this neighborhood. A simple and direct way is to form n vertex triples $\{T_l = (P, P_i, P_j) | P_i, P_j \in N_P, 1 \leq l \leq n\}$, and to consider curves interpolating each triple of vertices as the surface curves. Two issues arise here: one is how to choose two neighbor vertices P_i and P_j to form with P a vertex triple; another is which kind of curve will be used to interpolate each triple of vertices.

To choose the neighbor vertices P_i and P_j , we have to take into account the fact that the vertices in triangulation are sampling points of the real surface which are corrupted by noise. Since in equation (1) the function *cosine* is nonlinear, it is better to use the surface curves which are as close to the normal section as possible. In this way, the angle θ between the normal vector $\mathbf{n}$ of the surface curve and the surface normal $\mathbf{N}$ at P is close to 0 or π depending on the orientation of the surface normal, which falls in the low variation range of the function *cosine*, thus limiting the effects of the computation error for angle θ . On the other hand, the plane defined by two geometrically opposite vertices (with respect to P) and P is usually closer to the normal section than that defined by other combinations of vertices. These considerations lead to a simple strategy: we first define a quantity M to measure the geometric oppositeness of two neighbor vertices P_i and P_j as (see Fig. 2): $M = \langle P - P_i, P_j - P \rangle$. We then calculate this quantity for all combination of neighbor vertices, and sort them in a nonincreasing order. The first n combinations of the vertices are used to form n vertex triples with P . This strategy guarantees that n can always be greater than or equal to 3 which is the necessary condition for computing the principal curvatures by the Meusnier and the Euler theorem.

Having chosen the vertex triples, the next question is which kind of curve is to be used for their local interpolation. In our case, the only available information being P and its two neighbor vertices, their circumcircle appears to be the most stable curve which can be computed from these three vertices. Such a computation is also very efficient.

So we use the circles passing through each triple of vertices as an approximation of the surface curves.

Now suppose we have chosen a set of vertex triples $\{T_l | 1 \leq l \leq n\}$. Each vertex triple $T_l = (P, P_i, P_j)$ defines an intersecting plane to the underlying surface passing through P . The center C_l of the circumcircle of these three vertices can be easily computed [4]. Thus, the curvature and the unit normal vector of this circle at P are $\kappa_l = 1/||C_l - P||$ and $\mathbf{n}_l = (C_l - P)/||C_l - P||$, respectively. The unit tangent vector $\mathbf{t}_l$ at P is then

$$\mathbf{t}_l = \frac{\mathbf{n}_l \wedge (\mathbf{u} \wedge \mathbf{v})}{||\mathbf{n}_l \wedge (\mathbf{u} \wedge \mathbf{v})||}, \quad (7)$$

where $\mathbf{u} = P_i - P$ and $\mathbf{v} = P_j - P$. Hence for each triple T_l , we obtain for its circumcircle the curvature value κ_l , the tangent vector $\mathbf{t}_l$, and the normal vector $\mathbf{n}_l$ at P . We can then compute the surface normal $\mathbf{N}$ at P as

$$\mathbf{N} = \frac{\sum N_{mn}}{||\sum N_{mn}||}, \text{ where } N_{mn} = \frac{\mathbf{t}_m \wedge \mathbf{t}_n}{||\mathbf{t}_m \wedge \mathbf{t}_n||}, m \neq n. \quad (8)$$

Note that the orientation of each N_{mn} must be chosen to coincide with the choice of exterior (or interior) of the object (which can be decided from the surface triangulation).

Now, the normal curvature $\kappa_{\mathbf{t}_l}$ of the surface along the direction $\mathbf{t}_l$ can be obtained by equation (1) as:

$$\kappa_{\mathbf{t}_l} = \kappa_l \cos \theta, \quad (9)$$

where θ is the angle between $\mathbf{N}$ and $\mathbf{n}_l$. After choosing a coordinate system on the tangent plane passing through P , we can use equation (3) to compute a pair of coordinates (ξ_l, η_l) for each direction $\mathbf{t}_l$ and obtain an equation (5). Normally, n is greater than 3, so the three unknowns A , B , and C are often overdetermined. We can therefore use the least-squares technique to calculate them by minimizing the function:

$$G = \sum_1^n (A\xi_l^2 + 2B\xi_l\eta_l + C\eta_l^2 - \delta)^2, \quad (10)$$

where $\delta = \pm 1$ according to the orientation of the surface normal. The principal curvatures and directions can then be obtained as mentioned in 2.1.

3 Experimental Results

In order to characterize the performance of the method proposed above (TRI), we compare it with the method proposed by Besl and Jain (OP) [2] in which the second order orthogonal polynomials is used to approximate range data. This comparison is realized on a number of synthetic data sets. It is well known that even only with the truncation error inherent in data quantization, the curvature computation will be gravely deteriorated [2,5]. A common method to improve this is to first smooth the data by an appropriate Gaussian filter and then retain the results in floating-point form [5]. But the problem with such a smoothing is that it is often performed on the image coordinate system which is not intrinsic to the surface in image. The result is that the surface will be slightly modified which will lead to an incorrect curvature computation. So we will not use the smoothed version of the image in our comparison. We generate three synthetic range images of planar, spherical, and cylindrical surface with the following parameters:

Plane: $0.0125x - 0.0125y + 0.1f(x, y) = 1, \quad -30 \leq x, y \leq 30$.

Sphere: $x^2 + y^2 + f(x, y)^2 = 100, \quad -10 \leq x, y \leq 10, \quad f(x, y) \geq 0$.

Cylinder: $x^2 + f(x, y)^2 = 400$, $-20 \leq x, y \leq 20$, $f(x, y) \geq 0$.

For each kind of synthetic surface, we produce two data sets of different precision: one is obtained by sampling the surface at the grid points and retaining the results in floating-point form. This is referred to as the *32 bit image*. It is then truncated into 8-bit precision, which results in another data set called *8-bit image*. To obtain the surface triangulation, we apply a Delaunay triangulation-based surface approximation [7] to each image. Then both TRI and OP are applied to those vertices of the triangulation that are inside the synthetic surface. The window size of OP method is 7×7 .

We calculate two assessing measurements concerning these results: one is the mean μ of the computed $\hat{\kappa}_{max}$ and $\hat{\kappa}_{min}$, another is their standard deviation σ . They are listed in the following tables, with the true curvature values being in parenthesis.

Plane	$\hat{\kappa}_{max}$ (0.0)				$\hat{\kappa}_{min}$ (0.0)			
Image	μ -TRI	μ -OP	σ -TRI	σ -OP	μ -TRI	μ -OP	σ -TRI	σ -OP
32bit	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
8bit	0.000379	0.002125	0.000594	0.002162	-0.000489	-0.002348	0.000726	0.002239
Sphere	$\hat{\kappa}_{max}$ (0.1)				$\hat{\kappa}_{min}$ (0.1)			
Image	μ -TRI	μ -OP	σ -TRI	σ -OP	μ -TRI	μ -OP	σ -TRI	σ -OP
32bit	0.100000	0.100242	0.000000	0.000654	0.100000	0.100163	0.000000	0.000225
8bit	0.101514	0.108257	0.001898	0.005355	0.098985	0.097595	0.001645	0.022245
Cylinder	$\hat{\kappa}_{max}$ (0.05)				$\hat{\kappa}_{min}$ (0.0)			
Image	μ -TRI	μ -OP	σ -TRI	σ -OP	μ -TRI	μ -OP	σ -TRI	σ -OP
32bit	0.044019	0.050096	0.007106	0.000263	0.005353	0.000000	0.005911	0.000000
8bit	0.044352	0.052607	0.007384	0.009240	0.005011	0.000272	0.005845	0.001517

From these preliminary results, we see that TRI performs in generally better than OP. So it can be very useful for the applications where the surface triangulation is used. Other results can be found in [3], where we have also given an explanation of how to compute the Gaussian curvature without embedding the surface in a coordinate system.

References

- [1] Boissonnat, J.D.: Geometric structures for three-dimensional shape representation. ACM Trans. on Graphics, vol. 3, no. 4, pp. 266-286, 1984.
- [2] Besl, P.J, Jain, R.C.: Invariant surface characteristics for 3D object recognition in range image. Comput. Vision, Graphics, Image Processing, vol. 33, pp. 33-80, 1986.
- [3] Chen, X., Schmitt, F.: Intrinsic surface properties from surface triangulation. Internal report, Télécom Paris, Dec. 1991.
- [4] Faux, I.D., Pratt, M.J.: Computational geometry for design and manufacture. Ellis Horwood Publishers, 1979.
- [5] Flynn, P.J., Jain, A.K.: On reliable curvature estimation, in Proc. Conf. Computer Vision and Pattern Recognition, June 1989, pp. 110-116.
- [6] Lin, C., Perry, M.J.: Shape description using surface triangularization. in Proc. IEEE Workshop on Computer Vision: Repres. and Control, 1982, pp. 38-43.
- [7] Schmitt, F., Chen, X.: Fast segmentation of range images into planar regions. in Proc. IEEE Conf. on Computer Vision and Pattern Recognition, June 1991, pp.710-711.
- [8] Wolfson, E., Schwartz, E.L.: Computing minimal distances on polyhedral surfaces, IEEE Trans. Pattern Anal. Machine Intell., vol. 11, no. 9, pp. 1001-1005, 1989.